

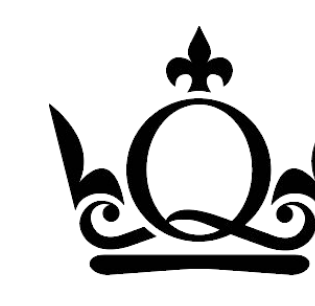
Latent Acoustic Mapping for Direction of Arrival Estimation

A Self-Supervised Approach

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I - Motivations

There are two essential challenges for acoustic mapping:

- Traditional models rely on ideal signal assumptions and can often suffer from performance limitations on real and noisy data.
- Compressed sensing methods bring significant improvements on quality and robustness, but rely on expensive iterative solvers.

Novel deep-learning methods for acoustic imaging:

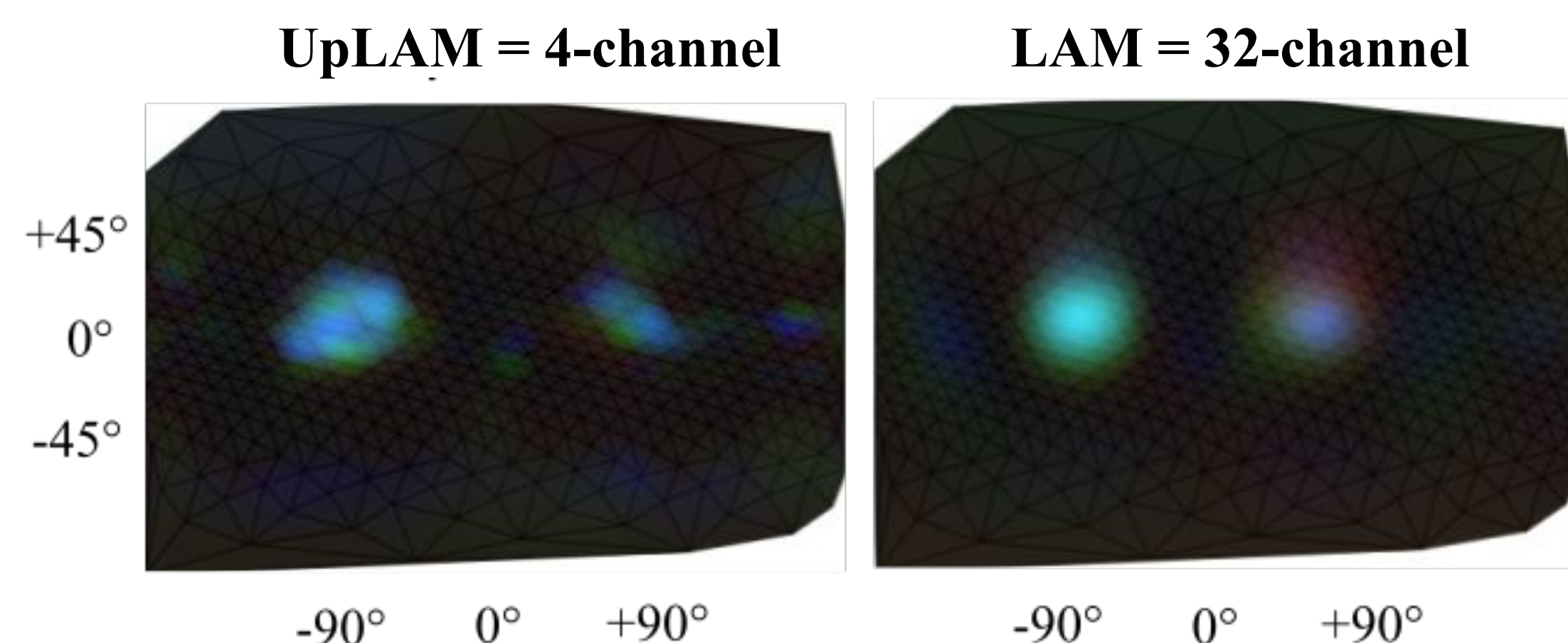
- DeepWave uses a graph convolutional architecture, unwrapping iterative processes into efficient feed-forward neural networks.
- However, DeepWave is supervised with ground truth maps from iterative solvers, which as a result bounds its performance.

II - Contributions

LAM: A self-supervised model for high resolution neural acoustic mapping from microphone array recordings.

- **Self-supervised:** optimizes directly the estimation of the input covariance matrix, bypassing the dependency on iterative solvers.
- **Data-driven:** adaptable and robust across low-and high-resolution microphone arrays and realistic acoustic events.
- **Impact:** LAM extends to low-resolution arrays via non-linear upsampling, and shows strong robustness on downstream DOA estimation, outperforming DeepWave and rivaling SOTA methods.

III - Acoustic Maps



Resources

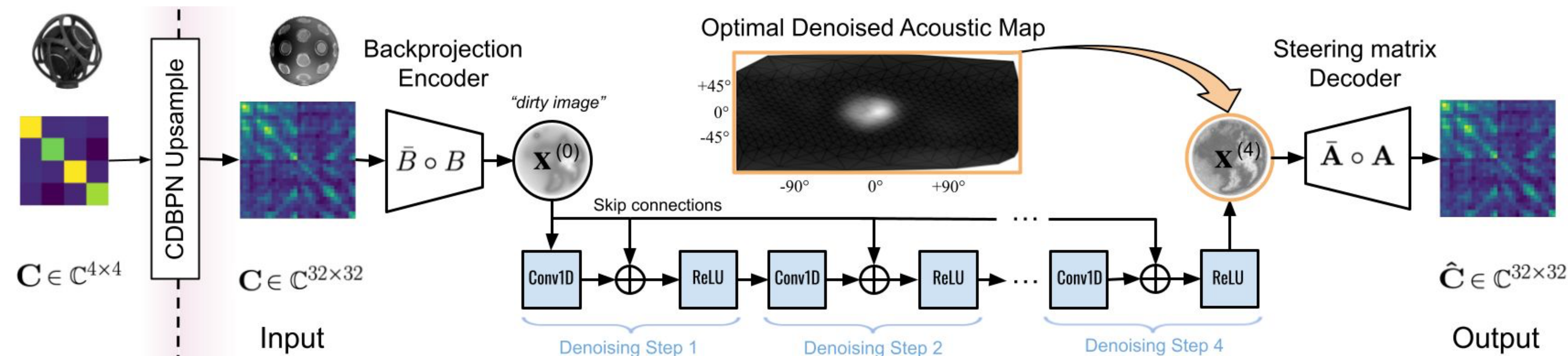
Research Paper



GitHub Repository



IV - LAM: Latent Acoustic Mapping



V - Architecture

Input: Cross Spectral Matrix

CSM computed from $y \in \mathbb{R}^{M \times T}$, M channels and duration T . The STFT of Y is used to compute the covariance at a k^{th} frequency band

$$\mathbf{C}_k = \frac{1}{N} \sum_{n=1}^N \mathbf{Y}(n, k) \mathbf{Y}(n, k)^H \in \mathbb{C}^{M \times M}$$

Encoder

A “back-projection” linear operation, maps $\mathbf{C}$ to a latent space $\mathbf{x}$. This is the first approximation to an acoustic map (i.e. “dirty image”)

$$\mathbf{x}^{(0)} = [\bar{\mathbf{B}} \circ \mathbf{B}]^H \text{vec}(\mathbf{C}) \in \mathbb{R}^N$$

Decoder

Given a steering matrix $\mathbf{A}$ for a mic array and a denoised latent $\mathbf{x}^{(4)}$, we approximate the original CSM

$$\mathbf{C} \approx \hat{\mathbf{C}} = \mathbf{A} \text{diag}(\mathbf{x}^{(4)}) \mathbf{A}^H = (\bar{\mathbf{A}} \circ \mathbf{A}) \mathbf{x}^{(4)}$$

Self-Supervised Pre-text Task

$$\mathbf{x}^{(4)} \in \arg \min_{\mathbf{x}^{(4)} \in \mathbb{R}_{\geq 0}^N} \left\| \mathbf{C} - \mathbf{A} \text{diag}(\mathbf{x}^{(4)}) \mathbf{A}^H \right\|_2^2$$

Loss function

$$L(\mathbf{C}, \hat{\mathbf{C}}) = \text{MSE}(\mathbf{C}, \hat{\mathbf{C}}) + \gamma \left(\left\| \mathbf{x}^{(4)} \right\|_1 + \sum_{i,j \in \mathcal{N}_i} |x_i^{(4)} - x_j^{(4)}| \right)$$

VI - Results

Input	Model	$LE \downarrow$	$LR \uparrow$
32ch	MUSIC→K-means	56.95°	97.2
	DeepWave→K-means [8]	14.8°	99.20
	LAM→K-means	13.69°	94.0
	LAM→GRU-MHSA	13.41°	94.7
4ch	MUSIC→K-means	61.83°	94.4
	DeepWave→K-means [8]	27.10°	99.20
	UpLAM→K-means	23.48°	97.8
	UpLAM→GRU-MHSA	14.44°	84.26
4ch	SELDnet [13]	16.8°	77.96
FOA	SELDnet [13]	22.43°	80.7
FOA+4ch	EINV2 [14]	19.83°	80.6

➤ Benchmark on the full LOCATA datasets.

Model	Input	$LE \downarrow$	$LR \uparrow$
MUSIC→GRU-MHSA	4ch	39.9°	73.5
DeepWave→GRU [8]	4ch	20.5°	70.1
UpLAM→GRU-MHSA	4ch	18.65°	57.6
SELDnet [13]	4ch	23.3°	82.3
SELDnet [13]	FOA	21.9°	83.0
EINV2 [14]	FOA+4ch	24.0°	84.2
UpLAM→GRU-MHSA †	4ch	18.31°	64.6
SELDnet [13] †	4ch	20.8°	66.4

➤ Benchmark on the STARSS “dev-test-sony” dataset split.